

Advanced (Carolina Reaper)

- Q1.** A bookshelf has 15 shelves, and each shelf can hold 18 books. If the bookshop owner wants to pack the books in boxes that hold 3 books each, how many boxes can be filled with 15×18 books?
(A) 90
(B) 135
(C) 150
(D) 180
(E) 270
- Q2.** What is the remainder when 43 is divided by 9?
(A) 0
(B) 1
(C) 2
(D) 4
(E) 7
- Q3.** If a bakery sells 252 loaves of bread per day, and they are packed in bags of 7 loaves each, how many bags are sold per day?
(A) 35
(B) 36
(C) 42
(D) 45
(E) 63
- Q4.** A group of friends want to share some candy equally. If they have 48 pieces of candy and there are 8 friends, how many pieces will each friend get, and what will be the remainder?
(A) 5 pieces, remainder 8
(B) 6 pieces, remainder 0
(C) 6 pieces, remainder 1
(D) 5 pieces, remainder 4
(E) 6 pieces, remainder 3
- Q5.** What is the remainder when 2^{10} is divided by 5?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4
- Q6.** A water tank can hold 240 gallons of water. If 3 buckets of water are poured into the tank per hour, and each bucket holds 5 gallons, what fraction of the tank will be filled after 8 hours?
(A) $\frac{1}{2}$
(B) $\frac{1}{3}$
(C) $\frac{2}{3}$
(D) $\frac{3}{4}$
(E) $\frac{4}{5}$
- Q7.** A clock strikes 6 times at 12 o'clock. If it strikes x times after y hours, find x when $y = 9$.
(A) 18
(B) 15
(C) 12
(D) 9
(E) 6
- Q8.** What is the remainder when 17 is divided by 5, and what is the quotient?
(A) Remainder: 2, Quotient: 3
(B) Remainder: 3, Quotient: 3
(C) Remainder: 2, Quotient: 4
(D) Remainder: 1, Quotient: 3
(E) Remainder: 2, Quotient: 2

Open Ended Questions (FRQ)

- Q9.** A person is planning a road trip that is 945 miles long. If they drive 5 hours per day at a constant speed of 45 miles per hour, what is the total number of days required to complete the trip? Show all work and calculations.
- Q10.** A bakery has 288 cupcakes to package for a large order. They want to put 8 cupcakes in each box. How many boxes are needed, and what is the remainder? Explain the reasoning behind the calculation.

Chef's Tips

- Ensure students have scratch paper for calculations
- Monitor students for adherence to the no-explanation rule
- Remind students to check their work carefully before submitting